

Lightweight Surface Defect Detection Algorithm Based on Improved YOLOv5

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Abstract—In response to the current issues of low detection accuracy and slow detection speed of detection algorithms in detecting surface defects on metal surfaces, a lightweight surface defect detection algorithm based on improved YOLOv5 is proposed. Firstly, a lightweight spatial attention module is introduced into the SPP module to extract effective information in the spatial dimension. Secondly, a new feature fusion network Amsf is designed to replace the original feature fusion network, which integrates feature information of different scales. Furthermore, the FAM attention mechanism is added to this network to enhance the detection capability for different types of targets. Finally, the SIoU loss function is used to replace the GIoU loss function to reduce boundary loss errors and improve the accuracy of target localization. Experimental results show that the improved model achieves an average detection accuracy of 81.97% on the NEU-DET dataset, which is 7.74% higher than YOLOv5. The model size and detection speed are comparable to YOLOv5. Effective improvement in detection accuracy is achieved while ensuring detection speed.

Keywords—surface defects; Defect detection; Loss function; YOLOv5; Attention mechanism

I. INTRODUCTION

In today's rapidly advancing technological era, smart manufacturing is gradually becoming the dominant trend in the manufacturing industry. Smart manufacturing utilizes advanced information technology, automation technology, and intelligent equipment to achieve automation and intelligence in the production process, thereby improving production efficiency. Defect detection plays a crucial role in smart manufacturing. Currently, surface defect detection methods mainly include manual inspection, non-destructive testing, and machine vision-based detection. Manual inspection suffers from issues such as fatigue, low efficiency, and subjectivity. Non-destructive testing methods include magnetic particle testing^[1], eddy current testing^[2], infrared testing^[3], ultrasonic testing^[4], etc. However, non-destructive testing methods have many limitations. Machine vision-based detection has become one of the solutions, with the increasing application of deep neural networks.

Currently, mainstream target detection algorithms represented by convolutional neural networks include Faster RCNN^[5], Mask R-CNN^[6], YOLO series^[7], SSD series^[8], etc. In practical applications, researchers continuously explore new algorithms and improve existing ones to enhance detection accuracy and speed. For example, Gao Jia et al^[9] proposed a network algorithm for detecting tensioned wire rope defects

(TWRD) based on the improved YOLOv5 network. By designing the LW-C3 module and the β -CIoU loss function, and improving the PAN structure, they balanced detection speed and accuracy to some extent in defect detection. Liu Yanju et al^[10] used the lightweight deep neural network MobileNetv3 to replace the feature extraction network CSPDarknet53 in YOLOv4 to improve detection speed. They also redefined confidence loss to address the problem of imbalanced positive and negative samples, effectively improving the accuracy of surface defect detection on hot-rolled steel bars. Yeung et al^[11] proposed a fusion attention network, FANet, for detecting various surface defects on steel, addressing issues such as surface defect shape changes and detection efficiency.

In summary, the current challenges in surface defect detection include low detection accuracy, which often leads to missed detections, and the dilemma of increasing model size while improving detection accuracy, resulting in slower detection speeds and difficulty deploying to mobile devices. To overcome these challenges, this paper proposes a lightweight surface defect detection algorithm based on improved YOLOv5. Firstly, the SPP-LSA module is used to replace the original SPP module, allowing for more effective utilization of information from multi-level pooled feature maps. Next, a new feature fusion network, Amsf, is designed to replace the original feature fusion network, with the addition of the FAM attention mechanism to fuse feature information of different scales. Finally, the SIoU loss function is employed to replace the GIoU loss function, enhancing the accuracy of defect localization. Through these improvements, the model's detection accuracy is enhanced, and detection speed is improved.

II. ALGORITHMIC MODEL

A. Overall network structure

The improved network structure, as shown in Figure 1, consists of three parts: Backbone, Neck, and Head. The Backbone performs feature extraction on the images, and the extracted features, referred to as feature layers, form the feature set of the input images. After passing through the Backbone, three feature layers C1, C2, and C3 are obtained. The Neck part is responsible for feature fusion of the three feature layers obtained in the main part, combining feature information of different scales. The Head part contains three detection heads, capable of obtaining information about objects of large, medium, and small sizes.

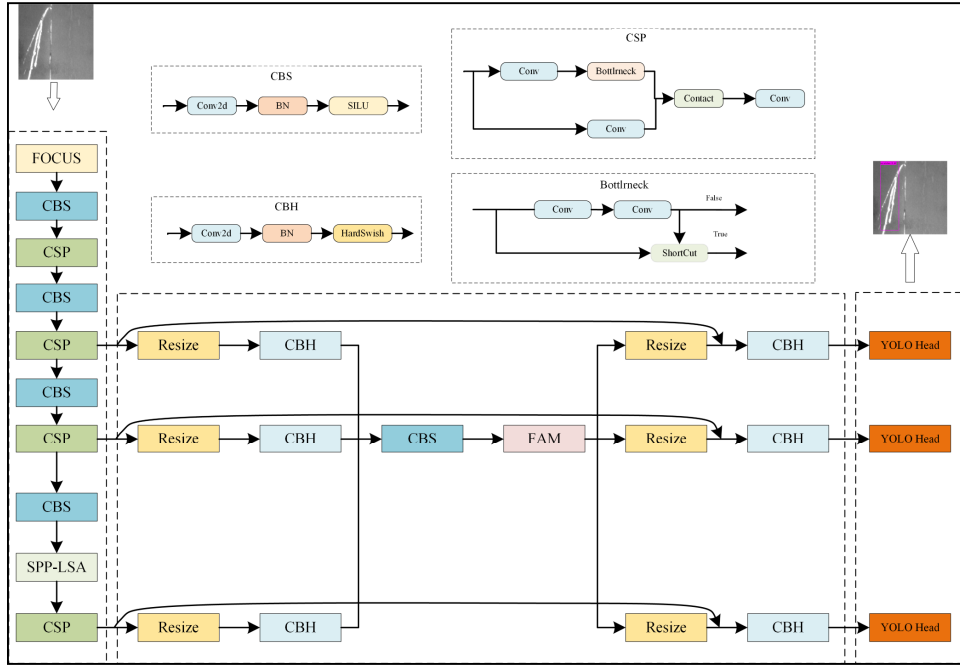


Figure 1 Network structure diagram

B. Amsf network structure

During the process of feature extraction for defects, three feature layers are obtained through the main network. Different scale feature maps contain different information; lower-level feature maps contain more information about small objects. As the information propagates from lower-level feature layers to higher-level ones through operations like convolution, larger objects undergo convolutional operations far more times than smaller objects. Therefore, the information of larger objects is more likely to be preserved in higher-level feature maps. Hence, it is necessary to construct a multi-scale fusion attention structure, Amsf (Attention multi-scale feature fusion), to merge feature maps of different scales, aggregate information from multiple levels, enrich feature map content, and enhance the detection capability of small defects on metal surfaces. The Amsf structure is depicted in Figure 2.

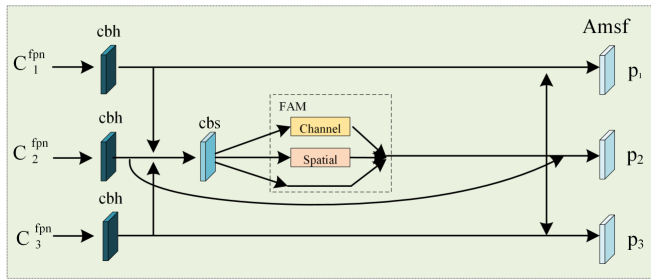


Figure 2 Amsf Structure Diagram

After the feature extraction of the backbone network, we can get the multi-level feature graphs C_1^{fpm} , C_2^{fpm} , C_3^{fpm} of defects. First, we adjust the multi-level feature layer to the size of the middle feature layer. Then, the resized feature map is passed through a 3×3 convolution kernel to generate three new feature maps, which are fused. The characteristic figure C_f is

formed. The expression of characteristic graph C_f is as follows:

$$C_f = \text{conv}_{3 \times 3} (\text{resize} (C_i^{fpm})) \quad (1)$$

In order to further enhance the fusion features, the convolution kernel of 3×3 is applied to the new feature map C_f to generate refined features. Then the attention mechanism is used to process the refinement feature. In the original panet, adding attention mechanism can only act on a single feature graph, but adding attention mechanism to multiple feature graphs will greatly increase the computational cost. For the Amsf network structure, multiple feature maps have been fused. Applying the attention mechanism to a single fused feature map can not only improve the accuracy of the attention mechanism, but also reduce a lot of computing costs. The C_{att} expression of the enhanced feature graph is as follows:

$$C_{att} = \text{att} (C_f) \quad (2)$$

Att means using the attention mechanism fam. Fam has two branches, channel branch and spatial branch, which are used to generate attention feature map. The fam structure is shown in Figure 3. Channel branching focuses on the representation of channels and meaningful content in images. The spatial branch focuses on spatial location and informativeness. In the spatial channel, the global average pooling (GAP) is used to process the input features. The input feature is converted into a channel information vector. Then, the channel information is further processed by multi-layer perceptron. In addition, the channel weights are normalized.

In spatial branching, the input features are first pooled to the maximum in the channel dimension to extract spatial information. Then, a 7×7 convolution operation is used to emphasize the context information. In addition, the spatial weights are normalized.

In order to enhance the feature map through the attention mechanism, the channel attention feature map is extended to the same size as the input feature map. Then, the expanded channel attention map is multiplied by the spatial attention map along

the channel dimension. In addition, the input features are multiplied by the integrated attention feature map to enhance the channel and spatial information. Get the output characteristic graph.

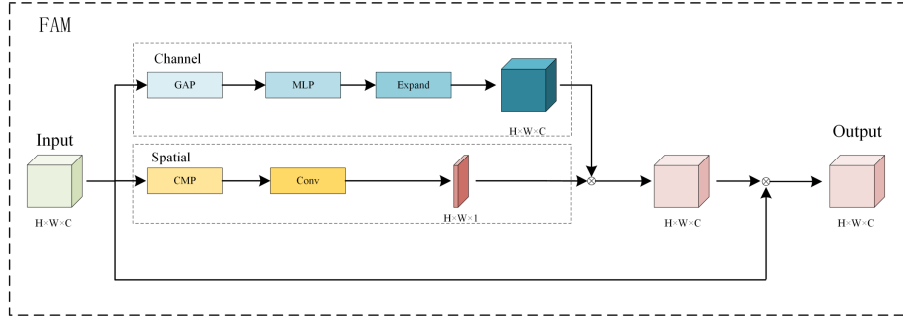


Figure 3 FAM structure diagram

The enhanced feature map needs to be resized to the original size of the multi-scale features to form multi-level features. In addition to multi-level features, the original multi-scale feature maps are added to the resized feature maps. This not only enhances the original features but also helps preserve feature information.

$$P_i = \sum_{i=\min}^{i=\max} \text{conv}_{3 \times 3} (\text{resize}(C_{att}) + C_i^{f_{pn}}) \quad (3)$$

Where P_i represents the output layer I feature map. Amsf network structure can fuse different levels of feature maps. This fusion method can extract the information of multi-layer features more balanced and enhance the information. In the detection of metal surface defects, it can better deal with different sizes and types of defects.

C. SPP-LSA module

To enhance the ability of the backbone network in channel information extraction, this paper proposes the SPP-LSA module, which replaces the SPP module in the YOLOv5s base network. This substitution is more flexible, with the main advantage being that it can achieve flexible and easy upgrades to the model through simple replacement without making significant changes to the overall network structure. The structure diagram of SPP-LSA is shown in Figure 4.

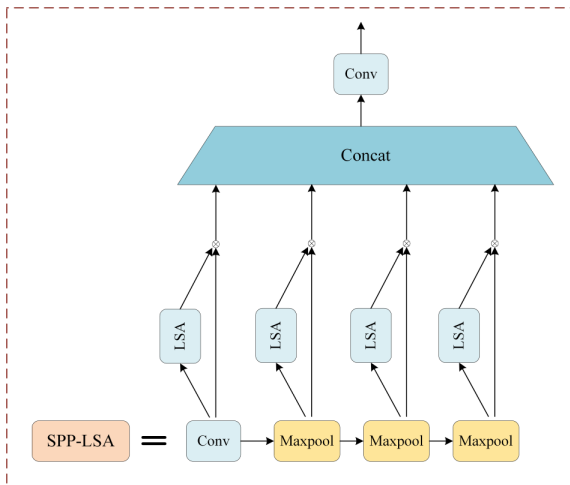


Figure 4 SPP-LSA structure

The LSA module utilizes global max-pooling (GMP) and global average-pooling (GAP) operations for acquiring global information. These operations, combined with parallel convolution operations, greatly enhance the module's performance. The LSA module first performs global max-pooling and global average-pooling operations to capture the most prominent features of each channel and the overall average features of the image, respectively. The outputs of these two operations enter parallel convolution operations, with one going through a standard 3x3 convolution and the other through a dilated convolution with a dilation rate of 1. This design of parallel convolution aims to effectively reduce computational burden while acquiring global spatial information. The structure of LSA is depicted in the figure 5.

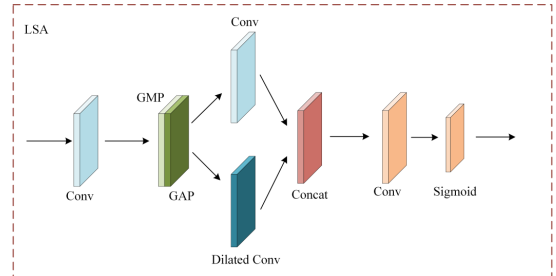


Figure 5 LSA structure diagram

After the parallel convolution operation, the two characteristic graphs are spliced in the channel dimension. This channel dimension stitching strategy is to improve the receptive field of the module and enable it to understand the structure of the image more comprehensively. Then, after a convolution operation, it is used to merge and restore the number of channels of feature importance. This step ensures that the feature map output by the module not only contains the global spatial information, but also maintains a moderate number of channels, thus providing a more optimized feature representation for the subsequent network layer integration. Through this structure, the LSA module, under the lightweight design, makes full use of the global information of GMP and GAP operations, extracts richer features through parallel convolution operations, and provides an effective and efficient solution for image spatial information processing of neural network.

D. IoU loss function

In loss calculation, the main calculation method of regression loss function is to use giou loss function, but giou loss function has the problem of inaccurate positioning of prediction frame. Siou loss function is used to replace giou loss function. SIOU optimizes the IoU by adding Angle loss to measure how similar the predicted box is to the real box, and through the gradient information of the similarity degree, it can more effectively guide the training of the model and improve the accuracy of detection and defect location. The Siou loss function mainly includes four parts: angle loss, distance loss, shape loss and IOU loss. The parameter diagram of Siou loss function is shown in Figure 6.

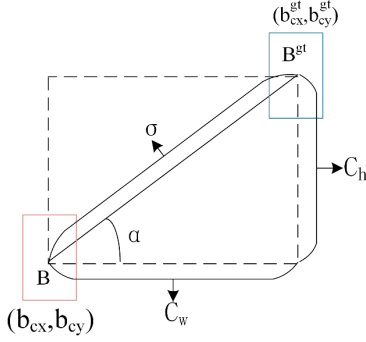


Figure 6 SIOU parameter diagram

In the figure 6, B and B^{gt} represent the centers of the predicted box and the ground truth box, where (b_{cx}, b_{cy}) and $(b_{cx}^{gt}, b_{cy}^{gt})$ are the central coordinates of the predicted and ground truth boxes, and C_w and C_h are the differences in the horizontal and vertical coordinates of the centers of the predicted and ground truth boxes. σ represents the distance between points B and B^{gt} , and α represents the angle between the line connecting the two points and the horizontal line.

Angle Loss Λ :

$$\Lambda = 1 - 2 * \sin^2 \left(\arcsin(x) - \frac{\pi}{4} \right) \quad (4)$$

In the angle loss:

$$x = \frac{C_h}{\sigma} = \sin(\alpha) \quad (5)$$

According to the definition of distance loss, the distance loss Δ is defined as follows:

$$\Delta = \sum_{t=x,y} (1 - e^{-\gamma \rho_t}) \quad (6)$$

Where ρ_x , ρ_y is given by:

$$\rho_x = \left(\frac{b_{cx}^{gt} - b_{cx}}{C_w} \right)^2, \quad \rho_y = \left(\frac{b_{cy}^{gt} - b_{cy}}{C_h} \right)^2, \quad \gamma = 2 - \Lambda \quad (7)$$

The definition of the shape loss is as follows:

$$\Omega = \sum_{t=w,h} (1 - e^{-\omega_t})^\theta \quad (8)$$

$$\omega_w = \frac{|w - w^{gt}|}{\max(w, w^{gt})} \quad \omega_h = \frac{|h - h^{gt}|}{\max(h, h^{gt})} \quad (9)$$

Where w and h denote the width and height of the ground truth box, and w^{gt} and h^{gt} are the width and height of the predicted box, respectively.

The expression for the SIOU loss function is as follows:

$$L = 1 - IOU + \frac{\Delta + \Omega}{2} \quad (10)$$

The SIOU loss introduces the concept of the angle between the predicted box and the ground truth box, assisting in the calculation of the distance between the two boxes. This accelerates the convergence of the network and brings the predicted box's position closer to the real position of the defect. It enhances the precision of defect localization, improving the model's ability to accurately locate defects.

III. EXPERIMENTAL RESULTS AND ANALYSIS

The experimental platform utilized in this study features an Intel Core i5-12400F processor, an NVIDIA RTX3060 GPU, and operates on the Windows 11 operating system. For deep learning framework, PyTorch is employed, leveraging CUDA 11.2 and cuDNN 8.1.1 to accelerate GPU computations. The optimizer utilized is Adam. Data augmentation is performed using Mosaic, and to prevent augmented images from deviating too far from the distribution of natural images, Mosaic augmentation is only applied during the first 70% of epochs.

A. Data set

This paper uses NEU-DET data set to evaluate the effectiveness and generality of the algorithm. Neu - det data set is a data set of metal plate surface defects released by Northeast University, which contains six types of steel surface defects, namely rolling chips (RS), plaques (PA), cracks (CR), dented surfaces (PS), inclusions (in) and scratches (SC). There are 300 samples of each type of defects, and a total of 1800 200*200 gray-scale images. We used the 8:2 ratio of training set and test set for training and testing.

B. Evaluation indicators

In this experiment, we used a variety of indicators to evaluate the performance of the detection model, including average precision and detection accuracy mAP@0.5 (mean average precision), recall, precision, parameters, and detection speed FPS (frames per second).

C. Experimental analysis

Table 1 presents the detection accuracy of the proposed method on the NEU-DET dataset, compared with other mainstream object detection models, including Faster RCNN, SSD, YOLOv4, YOLOv5, YOLOv7, and YOLOv8. As shown in Table 1, the LW-YOLO algorithm achieves an mAP of 81.97%, representing a 3.65% improvement compared to YOLOv8. Compared to YOLOv7, the mAP value is enhanced by 4.89%, demonstrating superior overall performance compared to other algorithms.

Among the seven algorithms listed in Table 1, LW-YOLO exhibits the highest detection accuracy for the Cr, In, and RS defect categories. Additionally, it demonstrates outstanding performance in detecting other defect categories. For the PS defect category, the detection accuracy is second only to YOLOv8.

TABLE 1 COMPARISON EXPERIMENTS OF MAINSTREAM DETECTION METHODS

Network	mAP(%)	AP(%)					
		Cr	In	Pa	PS	RS	Sc
YOLOv4	75.13	34.27	80.59	91.08	87.39	59.65	97.83
Faster RCNN	75.61	32.16	81.85	92.75	90.2	60.61	96.12
YOLOv5s	74.23	27.25	79.74	92.85	91.95	58.19	95.38
SSD	74.31	31.69	82.73	94.71	88.39	64.77	83.52
YOLOv7	77.08	36.06	83.37	94.73	91.35	63.14	93.81
YOLOv8	78.32	39.07	83.31	94.06	92.68	66.99	93.79
LW-YOLO	81.97	47.20	85.61	92.86	92.08	78.53	95.55

Table 2 displays the model parameters and detection speeds of each model. From Table 2, it can be observed that in defect detection, models such as YOLOv7, and YOLOv8 achieve higher mAP compared to YOLOv5s, although the difference is not significant. Additionally, in terms of model parameters, YOLOv7, and YOLOv8 have much larger parameter sizes than YOLOv5s. In defect detection, when the difference in detection accuracy is not significant, smaller models are more suitable for practical applications. This is also the reason why this study chose to modify YOLOv5s as the baseline model.

TABLE 2 COMPARISON OF MODEL PARAMETERS

Network	Params	FPS	mAP(%)
SSD	26.28	47.1	74.31
Faster RCNN	136.79	20.1	75.61
YOLOv4	63.97	38.1	75.13
YOLOv5s	7.16	64.2	74.23
YOLOv7	37.22	28.8	77.08
YOLOv8	11.13	53.2	78.32
LW-YOLO	7.1	70.2	81.97

After improvement upon YOLOv5s, LW-YOLO surpasses YOLOv5s, YOLOv7, and YOLOv8 in terms of both mAP and detection speed. The model proposed in this paper not only achieves faster detection speeds but also demonstrates an improvement in mAP values. In summary, the improved model exhibits higher performance in defect detection, achieving a better balance between model accuracy, frame rate, and model size compared to other models. It also possesses more comprehensive advantages in detection overall.

IV. CONCLUSION

This paper aims to solve the problem of surface defect detection accuracy and detection speed, and proposes a lightweight surface defect detection algorithm based on improved yolov5. Firstly, a lightweight spatial attention module is added to the SPP module to extract effective information from the spatial dimension. Secondly, a new feature fusion network Amsf is designed to replace the original feature fusion network and fuse different scales of feature information. The fam attention mechanism is added to the network to improve the detection ability for different types of targets. Finally, the Siou loss function is used to replace the giou loss function to reduce the boundary loss error. The experimental results show that the detection accuracy of the improved model on neu-det data set

is 81.97%. Compared with the current commonly used surface defect detection algorithm, it effectively improves the detection accuracy while maintaining a faster detection speed.

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